

Disaster Management System By Remotz Systems

Deliverable-1

Project Team

Muhammad Moiz Ansari	23i-0523
Zohaib Umer	23i-0528
Rehan Tajik	23i-0730
Muhammad Tayyab	23i-0738

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Supervised by
Ms. Rabail Zahid



**Department of Computer Science National University of
Computer and Emerging Sciences Islamabad, Pakistan**

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PART C - Deliverable 1:

1. Introduction

The Disaster Management System is a comprehensive software solution designed to streamline coordination between citizens, NGOs, rescue teams, and administrators during crisis situations. By digitizing the reporting and response process, the system aims to minimize response times and ensure resources are allocated efficiently to areas in need.

2. Project Vision

To create a centralized, resilient platform that empowers communities and response organizations to collaborate effectively during disasters, ultimately saving lives and reducing property damage through timely information and organized action.

3. Intended Use of the System

Stakeholders and Needs

- **Citizens (Main Beneficiaries):**
 - **Need:** To quickly report disasters, request urgent assistance, and receive real-time updates on safety status.
 - **Usage:** They will use the mobile/web interface to submit reports and track their requests.
- **NGOs & Rescue Teams:**
 - **Need:** To view a verified list of disasters, accept rescue missions, and provide feedback on the ground situation.
 - **Usage:** They will log in to view assigned tasks, update status, and coordinate with the admin.
- **Admin:**
 - **Need:** Centralized control to verify incidents, mobilize resources (assign NGOs/Rescue teams), and analyze post-disaster data.
 - **Usage:** They will use the dashboard to manage users, records, and generate reports.

4. Features / Overall Functionality

- **User Management:** Secure registration and authentication for different roles (Admin, NGO, Citizen).
- **Disaster Reporting:** A streamlined interface for citizens to report incidents with location and severity details.

- **Resource Allocation:** Functionality for Admins to assign specific NGOs or Rescue Teams to reported incidents.
- **Alert System:** Real-time broadcasting of alerts to stakeholders regarding new disasters or status updates.
- **Record Management:** CRUD (Create, Read, Update, Delete) capabilities for managing disaster records and historical data.

5. User Stories

The following user stories define the functional requirements for Sprint 1:

- **US-01:** Register User
- **US-02:** Manage Disaster Records (Epic)
 - **US-03:** Add Disaster Record
 - **US-04:** Edit Disaster Record
 - **US-05:** Delete Disaster Record
- **US-06:** Report Disaster
- **US-07:** Send Notification/Alert (Epic)
 - **US-08:** Broadcast Alert to Stakeholders
 - **US-09:** Send Notification
- **US-10:** View Disaster Alerts

6. Structured Specifications

Note: Below are the specifications for the sub-user stories and independent stories implemented in this sprint.

US-01: Register User

- **Description:** As a user (Citizen/NGO), I want to create an account so that I can access the system features.
- **Pre-condition:** User is on the registration page.
- **Post-condition:** New user account is stored in the database.
- **Acceptance Criteria:**
 1. System validates email format and password strength.
 2. User receives a success message upon valid submission.
 3. System prevents duplicate email registration.

US-03: Add Disaster Record (Sub-story of US-02)

- **Description:** As an Admin, I want to add a new disaster record so that it is visible to rescue teams.
- **Pre-condition:** Admin is logged in.

- **Post-condition:** Disaster record is added to the central database.
- **Acceptance Criteria:**
 1. Admin can enter details (Type, Location, Severity, Date).
 2. The system generates a unique ID for the disaster.
 3. The record appears immediately in the dashboard list.

US-04: Edit Disaster Record (Sub-story of US-02)

- **Description:** As an Admin, I want to update disaster details so that information remains accurate.
- **Pre-condition:** Admin is logged in, and a record exists.
- **Post-condition:** Record is updated in the database.
- **Acceptance Criteria:**
 1. Admin can modify the status (e.g., Active to Resolved).
 2. System saves changes and logs the timestamp of the update.

US-05: Delete Disaster Record (Sub-story of US-02)

- **Description:** As an Admin, I want to remove invalid records so that the system remains clutter-free.
- **Pre-condition:** Admin is logged in.
- **Post-condition:** Record is removed or flagged as deleted.
- **Acceptance Criteria:**
 1. The system asks for confirmation before deletion.
 2. The deleted record is no longer visible in the active list.

US-06: Report Disaster

- **Description:** As a Citizen, I want to report a disaster so that rescue teams can be notified.
- **Pre-condition:** Citizen is logged in.
- **Post-condition:** A pending disaster report is sent to the Admin for verification.
- **Acceptance Criteria:**
 1. User must provide mandatory fields (Location, Type).
 2. System confirms submission with a reference number.

US-08: Broadcast Alert to Stakeholders (Sub-story of US-07)

- **Description:** As an Admin, I want to broadcast an alert so that all NGOs are aware of a critical situation.
- **Pre-condition:** A verified disaster exists.
- **Post-condition:** Alert is sent to all registered NGOs/Rescue teams.
- **Acceptance Criteria:**

1. The message is delivered to all target users in the "NGO" group.
2. System displays a "Broadcast Sent" confirmation.

US-09: Send Notification (Sub-story of US-07)

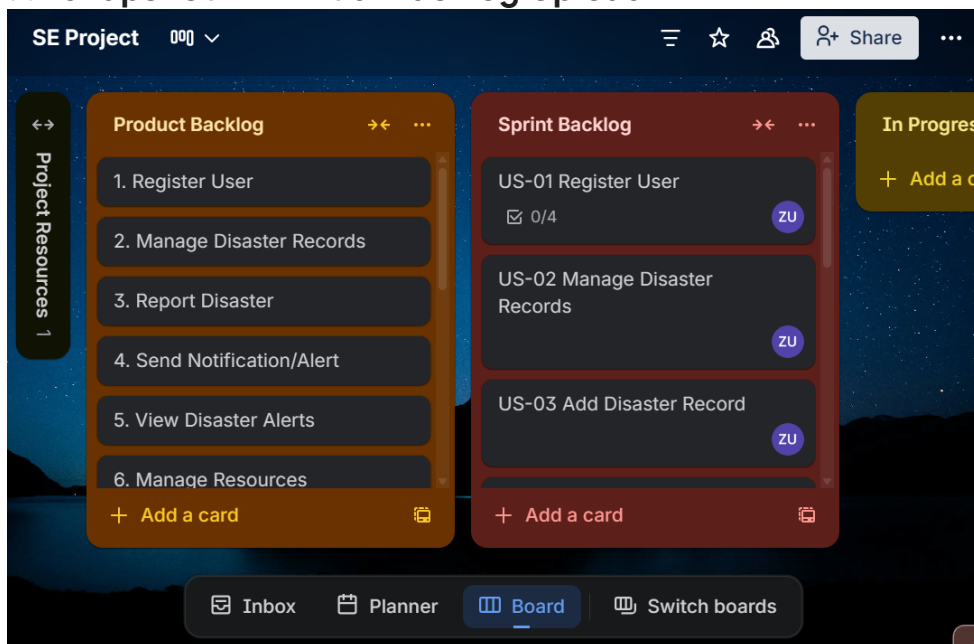
- **Description:** As the System, I want to send a notification to a specific user so they know their request status changed.
- **Pre-condition:** The status of a request changes.
- **Post-condition:** User receives a specific notification.
- **Acceptance Criteria:**
 1. Notification appears in the user's inbox/dashboard.
 2. Notification content matches the specific status update.

US-10: View Disaster Alerts

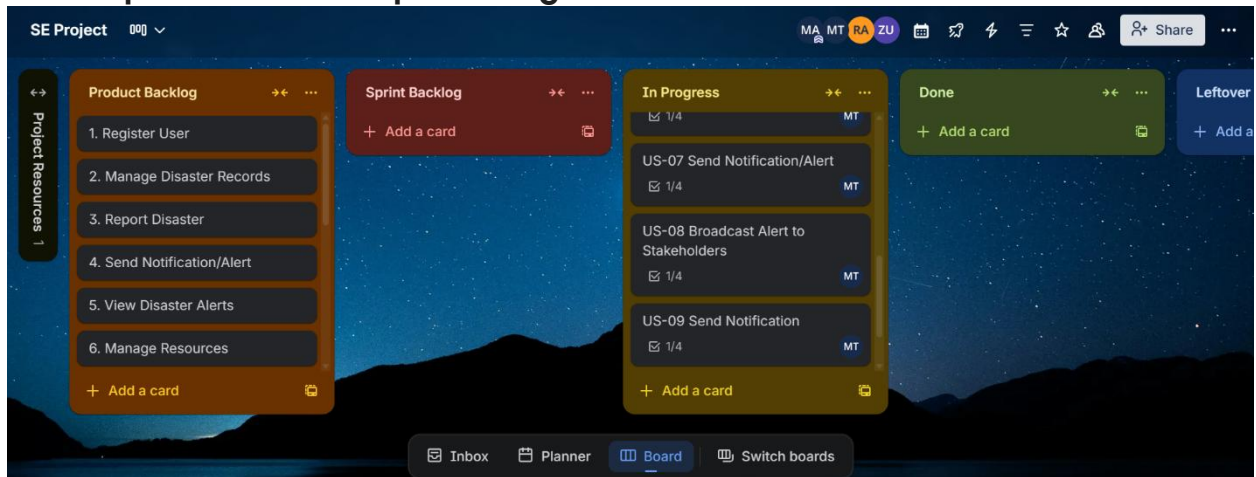
- **Description:** As an NGO/Citizen, I want to view alerts so that I can stay informed about safety.
- **Pre-condition:** User is logged in.
- **Post-condition:** User sees a list of active alerts.
- **Acceptance Criteria:**
 1. Alerts are sorted by date/time (newest first).
 2. The user can filter alerts by location.

7. Scrum Board

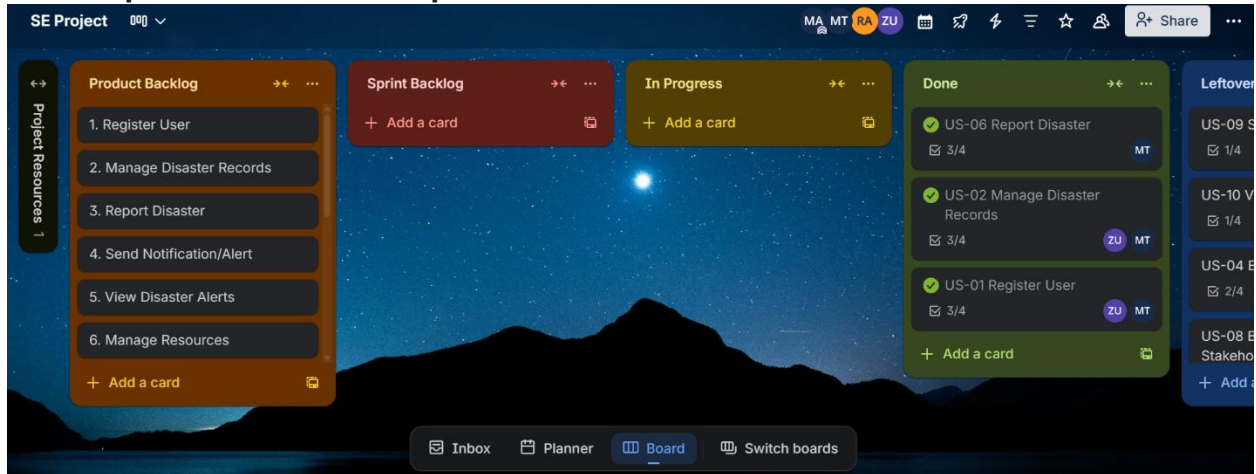
7.1 Snapshot 1 — Initial Backlog Upload



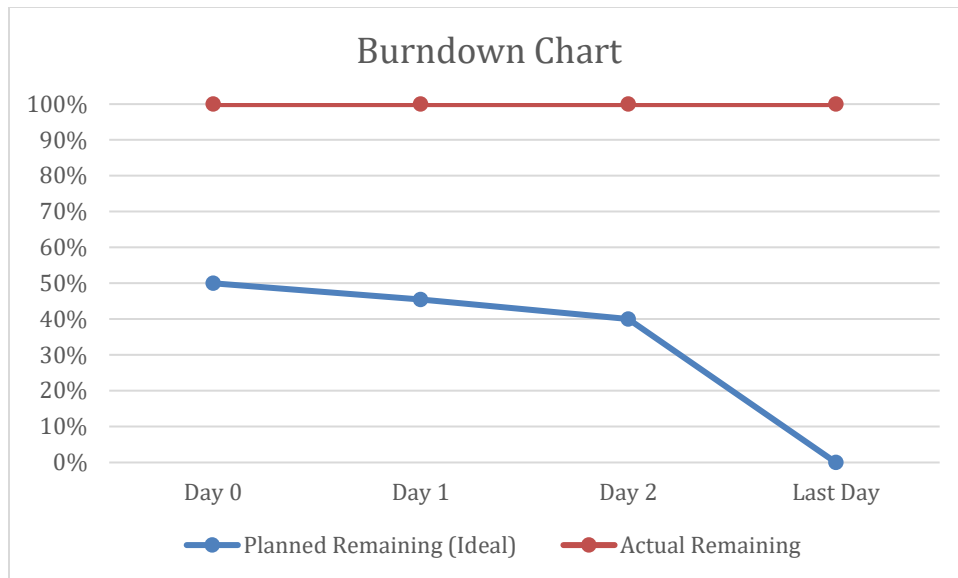
7.2 Snapshot 2 — Mid-Sprint Progress



7.3 Snapshot 3 — End of Sprint



7.4 Product Burn-Down Chart — Sprint 1



8. NFR Specification (Natural Language)

- **Performance:** The system shall send broadcast alerts to all stakeholders within 5 seconds of the Admin initiating the alert.
- **Availability:** The system must be available 99.9% of the time to ensure citizens can report disasters at any moment.
- **Security:** Only users with "Admin" role privileges can access the "Manage Disaster Records" and "Broadcast Alert" modules; passwords must be encrypted in the database.
- **Usability:** The "Report Disaster" form should be navigable and submittable within 3 clicks to ensure ease of use during panic situations.

9. Iteration 1 Implementation Screenshots

9.1 Login Page

Disaster Management System - Login



Disaster Management System

Coordinating Response. Saving Lives.

Emergency Alert
*** Report disasters immediately to enable rapid response

Welcome Back

Sign in to your account to continue

Sign In

Sign Up

Login as

Admin

Username

Enter your username

Password

Enter your password

Forgot password?

Emergency Hotline: **1122**

9.2 Signup/Register Page

Disaster Management System - Login



Disaster Management System

Coordinating Response. Saving Lives.

Emergency Alert
*** Report disasters immediately to enable rapid response

Welcome Back

Sign in to your account to continue

Sign In

Sign Up

Register as

Citizen

Name *

Enter your full name

CNIC *

Enter CNIC (e.g., 12345-1234567-1)

Contact Number *

Enter phone number

Emergency Hotline: **1122**

9.3 Manage Disaster Page

Manage Disasters

View and manage all disaster records

Total Disasters

17

Active

11

Contained

4

Resolved

2

🔍 Search disasters...

All Types

All Statuses

+ Add Disaster

ID	Type	Location	Severity	Status	Date	Team	NGO
DIS022	Fire	Islambad	Critical	Active	Jan 15, 2026 10:24 ...	Unassigned	Unassigned
DIS019	Flood	Saddar, Karachi	High	Active	Jan 15, 2026 09:04 ...	Unassigned	Unassigned
DIS020	Earthquake	Kashmit	High	Under Control	Jan 15, 2026 12:00 ...	The Avengers	Edhi Foundation
DIS018	Earthquake	Kashmir	High	Active	Dec 3, 2025 12:00 ...	Unassigned	UNICEF Pakistan
DIS013	Flood	Village, Karachi	High	Active	Nov 23, 2025 09:38...	Unassigned	Unassigned

10. Work Division

- **Moiz Ansari:** Analyzed and figured out the required user stories. Divided tasks among members. Created team information document and created base document for sprint 1. Created a Trello board. Created github repository. Provided UI design for the system as well.
 - **Rehan Tajik:** Created Deliverable Document.
 - **Zohaib Umer:** Developed US-01 to US-05 and sub-stories (CRUD operations).
 - **Tayyab:** Developed US-06 to US-10 and sub-stories (CRUD operations).
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